

MTH 204 : Lecture 03

Ex: $[\cos y \sinh(x) + 1] dx - [\sin y \cosh(x)] dy = 0$
given $y(1) = 2$

Note: $\sinh(x) = \frac{e^x - e^{-x}}{2}$ and $\cosh(x) = \frac{e^x + e^{-x}}{2}$

Then $\frac{d[\sinh(x)]}{dx} = \frac{e^x + e^{-x}}{2} = \cosh(x)$ and $\frac{d[\cosh(x)]}{dx} = \frac{e^x - e^{-x}}{2} = \sinh(x)$

Now $M(x, y) = \cos y \sinh(x) + 1$

$$N(x, y) = -[\sin y \cosh(x)]$$

$$\frac{\partial M}{\partial y} = -\sin y \sinh(x) \text{ and } \frac{\partial N}{\partial x} = -\sin y \sinh(x)$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{The differential equation is exact.}$$

Now $\frac{\partial u}{\partial x} = M(x, y) = \cos y \sinh(x) + 1$

Then integrating with respect to x ,

$$u(x, y) = \cos y \cosh(x) + x + g(y)$$

Now $\frac{\partial u}{\partial y} = -\sin y \cosh(x) + g'(y) = N(x, y)$

Therefore $-\sin y \cosh(x) + g'(y) = -\sin y \cosh(x)$

$$\Rightarrow g'(y) = 0 \Rightarrow g(y) = C_1 (\text{Constant})$$

Thus $u(x, y) = \cos y \cosh(x) + x + C_1$

So, the general solution of the ODE is $u(x, y) = C_2$

$$\Rightarrow \boxed{\cos y \cosh(x) + x = C} \quad (C = C_2 - C_1)$$

$$\text{Now } y(1) = 2 \Rightarrow \cos 2 \cos h(1) + 1 = C$$

So, the particular solution of the IVP is:

$$\boxed{\cos y + \cos h(x) + x = \cos 2 \cos h(1) + 1}$$

Ex: $-y dx + x dy = 0$

Here $M = -y$, $N = x$

$$\frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1$$

So, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ and so the above ODE is not exact.

However it is separable. $-y dx + x dy = 0$
 $\Rightarrow \frac{dy}{y} = \frac{dx}{x}$

$$\Rightarrow \ln|y| = \ln|x| + C, \Rightarrow \boxed{y = Cx}$$

Note: Although the above ODE is not exact, it becomes exact if we multiply it by $\frac{1}{x^2}$

$$-y dx + x dy = 0 \Rightarrow -\frac{y}{x^2} dx + \frac{1}{x} dy = 0$$

$$\Rightarrow d\left(\frac{y}{x}\right) = 0 \Rightarrow \frac{y}{x} = C \Rightarrow \boxed{y = Cx}$$

Since $\frac{\partial(-y/x^2)}{\partial y} = -\frac{1}{x^2}$ and $\frac{\partial(1/x)}{\partial x} = -\frac{1}{x^2}$,

the above equation is an exact ODE and

can be written as $d\left(\frac{y}{x}\right) = 0$

$$\Rightarrow \frac{y}{x} = c$$

- can also multiply by $\frac{1}{y^2}$

Then $\frac{1}{y^2}(-y dx + x dy) = -\frac{1}{y} dx + \frac{x}{y^2} dy = 0$

$$\Rightarrow d\left(\frac{x}{y}\right) = 0$$

$$\Rightarrow \frac{x}{y} = \text{constant}$$

- Can also multiply by $\frac{1}{xy}$

Then $\frac{1}{xy}(-y dx + x dy) = -\frac{1}{x} dx + \frac{1}{y} dy = 0$

$$\Rightarrow d\left(\ln\left(\frac{y}{x}\right)\right) = 0$$

- Can also multiply by $\frac{1}{x^2+y^2}$

$$\text{Then } \frac{1}{x^2+y^2} (-y dx + x dy) = 0 \Rightarrow -\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy = 0$$

$$\Rightarrow d\left(\tan^{-1} \frac{y}{x}\right) = 0$$

Definition: An integrating factor I.F. is a function

$F(x, y)$ such that the equation

$$M(x, y) dx + N(x, y) dy = 0$$

becomes an exact ODE after multiplication:

i.e. $FM dx + FN dy = 0$ is an exact equation.

- I.F. of an ODE is not unique.
- A non exact ODE may not have an I.F.

How to Find I.F. :

$$\text{Let } M dx + N dy = 0$$

Multiplying by $F(x, y)$ we have

$$F(x, y) M(x, y) dx + F(x, y) N(x, y) dy = 0$$

This ODE will be exact if

$$\frac{\partial(FM)}{\partial y} = \frac{\partial(FN)}{\partial x}$$

$$M \frac{\partial F}{\partial y} + F \frac{\partial M}{\partial y} = N \frac{\partial F}{\partial x} + F \frac{\partial N}{\partial x}$$

- If $M dx + N dy = 0$ is already exact,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Then F is an I.F. if $\boxed{M \frac{\partial F}{\partial y} = N \frac{\partial F}{\partial x}}$

Otherwise

after multiplication the ODE is exact

$$\text{if } M \frac{\partial F}{\partial y} + F \frac{\partial M}{\partial y} = N \frac{\partial F}{\partial x} + F \frac{\partial N}{\partial x}$$

- If the I.F. F depends only on x and not on y , then $\frac{\partial F}{\partial y} = 0$

$$\text{So, } F \frac{\partial M}{\partial y} = N \frac{dF}{dx} + F \frac{\partial N}{\partial x}$$

Dividing both sides by FN ,

$$\frac{1}{N} \frac{\partial M}{\partial y} = \frac{1}{F} \frac{dF}{dx} + \frac{1}{N} \frac{\partial N}{\partial x}$$

$$\Rightarrow \frac{1}{F} \frac{dF}{dx} = \frac{1}{N} \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial x} \right)$$

$$\Rightarrow \int \frac{dF}{F} = \int \frac{1}{N} \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial x} \right) dx$$

$$\Rightarrow \boxed{F = e^{\int \frac{1}{N} \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial x} \right) dx}}$$

- If the I.F. F depends only on y and not on x (i.e. $\frac{\partial F}{\partial x} = 0$)

then $M \frac{dF}{dy} + F \frac{\partial M}{\partial y} = F \frac{\partial N}{\partial x}$

$$\Rightarrow \frac{1}{F} \frac{dF}{dy} + \frac{1}{M} \frac{\partial M}{\partial y} = \frac{1}{M} \frac{\partial N}{\partial x}$$

$$\Rightarrow \frac{1}{F} \frac{dF}{dy} = \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial x} \right)$$

$$\Rightarrow \int \frac{dF}{F} = \int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial x} \right) dy$$

$$\Rightarrow \boxed{F = e^{\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial x} \right) dy}}$$

Ex:
$$\underbrace{(e^{x+y} + ye^y)}_M dx + \underbrace{(xe^y - 1)}_N dy = 0$$

check for exactness:

$$\frac{\partial M}{\partial y} = \frac{\partial (e^{x+y} + ye^y)}{\partial y} = e^{x+y} + ye^y + e^y$$

$$\frac{\partial N}{\partial x} = \frac{\partial (xe^y - 1)}{\partial x} = e^y \neq \frac{\partial M}{\partial y}$$

So, this ODE is not exact

$$\text{Now } \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = e^{x+y} + ye^y + e^y - e^y = e^{x+y} + ye^y$$

$$\Rightarrow \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = \frac{-\cancel{(e^{x+y} + ye^y)}}{\cancel{(e^{x+y} + ye^y)}} = -1$$

So, the ODE has an I.F. depending on y only.

$$\begin{aligned} \text{Hence I.F. } F &= e^{\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy} \\ &= e^{-\int dy} = e^{-y} \end{aligned}$$

Multiplying the original ODE by e^{-y} ,

$$(e^x + y) dx + (x - e^{-y}) dy = 0$$

check: $\frac{\partial(e^x + y)}{\partial y} = 1$, $\frac{\partial(x - e^{-y})}{\partial x} = 1$

So, this is an exact equation.

So, there exists $u(x, y)$ such that

$$\frac{\partial u}{\partial x} = e^x + y \quad , \quad \frac{\partial u}{\partial y} = x - e^{-y}$$

$$u(x) = \int (e^x + y) dx = e^x + yx + g(y)$$

$$\frac{\partial u}{\partial y} = x + g'(y) = x - e^{-y}$$

$$\Rightarrow g'(y) = -e^{-y} \Rightarrow g(y) = e^{-y} + \text{constant}$$

So, the general solution of the ODE is

$$u(x, y) = e^x + yx + e^{-y} = C$$

Ex:
$$\underbrace{3(y+1)}_M dx - \underbrace{2x}_{N} dy = 0$$

$$\frac{\partial M}{\partial y} = 3, \quad \frac{\partial N}{\partial x} = -2$$

So, this ODE is not exact.

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{3 - (-2)}{-2x} = -\frac{5}{2x}$$

This is a function of x only.

$$\begin{aligned} \text{IF } F &= e^{\int -\frac{5}{2x} dx} = e^{-5/2 \ln x} = e^{\ln(x^{-5/2})} \\ F &= x^{-5/2} \end{aligned}$$

Multiplying the ODE by $x^{-5/2}$ we get

$$3x^{-5/2}(y+1) dx - 2x^{-3/2} dy = 0$$

Check:
$$\frac{\partial [3x^{-5/2}(y+1)]}{\partial y} = 3x^{-5/2}$$

$$\frac{\partial (-2x^{-3/2})}{\partial x} = (-2)\left(-\frac{3}{2}\right)x^{-3/2-1} = 3x^{-5/2}$$

So, this new ODE is exact.

Thus there exists a function $u(x, y)$ such that

$$\frac{\partial u}{\partial x} = 3x^{-5/2}(y+1)$$

$$u(x, y) = \frac{3x^{-5/2+1}}{(-5/2+1)}(y+1) + g(y)$$

$$= \frac{3x^{-3/2}}{(-\frac{3}{2})}(y+1) + g(y)$$

$$= -2x^{-3/2}(y+1) + g(y)$$

$$\frac{\partial u}{\partial y} = -2x^{-3/2} + g'(y) = -2x^{-3/2}$$

$$\Rightarrow g'(y) = 0 \Rightarrow g(y) = C_1$$

So, the solution is $u(x, y) = C_2$

$$\Rightarrow -2x^{-3/2}(y+1) + C_1 = C_2 \Rightarrow -2x^{-3/2}(y+1) = C_2 - C_1$$

$$\Rightarrow 2x^{-3/2}(y+1) = C_1 - C_2 \Rightarrow y+1 = \frac{(C_1 - C_2)}{2} x^{3/2}$$

$$\Rightarrow y+1 = C x^{3/2} \quad \text{where } C = \frac{C_1 - C_2}{2}$$

$$\Rightarrow \boxed{y = Cx^{3/2} - 1}$$

Table for IF

$$\bullet \quad \frac{M_y - N_x}{N} = f(x) \Rightarrow \text{IF: } F = e^{\int f(x) dx}$$

$$\bullet \quad \frac{N_x - M_y}{M} = g(y) \Rightarrow \text{IF: } F = e^{\int g(y) dy}$$

(Some other cases)

$$\bullet \quad \frac{M_y - N_x}{yN - xM} = h(xy), \text{ I.F. } F = e^{\int h(z) dz} \text{ where } z = xy$$

⋮

Linear ODE: A first order ODE is

called Linear if it can be written as:

$$y' + p(x)y = r(x).$$

(It is a linear combination of y' , y and 1
Coefficients/scalars are functions of x only)

Note that $a(x) y' + b(x) y + c(x) = 0$

$$\Rightarrow y' + \underbrace{\frac{b(x)}{a(x)}}_{p(x)} y = - \underbrace{\frac{c(x)}{a(x)}}_{r(x)}$$

Homogeneous Linear ODE:

A linear, first order ODE is said to be homogeneous if $r(x) = 0$ (In the interval in consideration)
and hence $y' + p(x)y = 0$

Separation of Variables:

$$\frac{dy}{y} = -p(x)dx$$
$$\Rightarrow \boxed{y = c e^{-\int p(x)dx}}$$

$y=0$ is also a solution (Trivial solution)

Non homogeneous Linear ODE:

$$y' + p(x)y = r(x)$$

$$\Rightarrow (p(x)y - r(x))dx + dy = 0$$

Then $\frac{\partial M}{\partial y} = p(x)$, $\frac{\partial N}{\partial x} = 0$ (This is nonexact if $p(x) \neq 0$)

$$\frac{M_y - N_x}{N} = \frac{p(x) - 0}{1} = p(x) \quad \left(\begin{array}{l} \text{a function} \\ \text{of } x \text{ only} \end{array} \right)$$

I.F. $F = e^{\int p(x) dx}$

Multiplying by the I.F.,

$$e^{\int p(x) dx} y' + e^{\int p(x) dx} p(x)y = e^{\int p(x) dx} r(x)$$

$$\Rightarrow \frac{d \left(y e^{\int p(x) dx} \right)}{dx} = r(x) e^{\int p(x) dx}$$

$$\Rightarrow y e^{\int p(x) dx} = \int r(x) e^{\int p(x) dx} dx + C$$

$$\Rightarrow y = e^{-\int p(x) dx} \left[\int r(x) e^{\int p(x) dx} dx + C \right]$$

Ex: $y' + y \tan x = e^{-.01x} \cos x$, $y(0) = 0$

$$p(x) = \tan x , \quad r(x) = e^{-.01x} \cos x$$

$$\int p(x) dx = \int \tan x dx = \log |\sec x|$$

$$\text{I.F.} = e^{\int p(x) dx} = |\sec x|$$

$$e^{-\int p(x) dx} = |\cos x|$$

Near $x=0$, $\cos x > 0$ and $|\cos x| = \cos x$

$$\text{So, } y(x) = \cos x \left(\int \sec x e^{-.01x} \cos x dx + C \right)$$

$$= \cos x \left[\int e^{-.01x} dx + C \right]$$

$$= \cos x \left[\frac{e^{-.01x}}{-.01} + C \right] = \cos x \left[-100 e^{-.01x} + C \right]$$

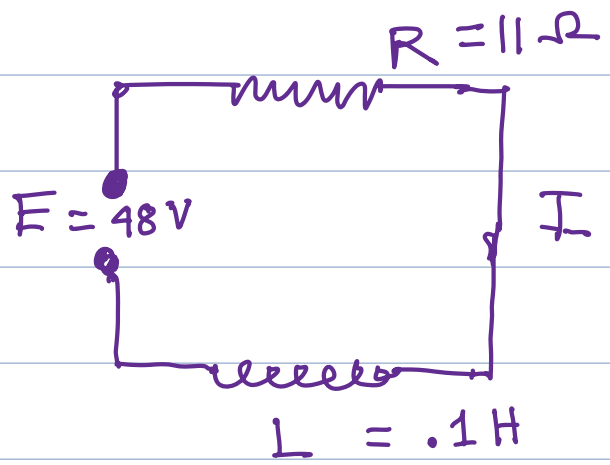
$$\text{Now } y(0) = 0 \Rightarrow 0 = 1 [-100 + C] \Rightarrow C = 100$$

Then $y(x) = 100 \cos x [1 - e^{-.01x}]$

Ex:

Voltage drop
across the resistor
 $= IR$

Voltage drop across
inductor $= LI' = L \frac{dI}{dt}$



(LR circuit)

$$E = LI' + IR$$

$$\Rightarrow \boxed{I' + \frac{R}{L}I = \frac{E}{L}} \quad (\text{Linear ODE})$$

$$p(t) = \frac{R}{L}$$

$$\int p(t) dt = \int \frac{R}{L} dt = \frac{R}{L} t$$

$$\text{So, } I(t) = e^{-\frac{R}{L}t} \left[\int e^{\frac{R}{L}t} \frac{E}{L} dt + C \right]$$

$$= e^{-\frac{R}{L}t} \left[\cancel{\frac{E}{L}} \cdot \frac{e^{\frac{R}{L}t}}{\cancel{\frac{R}{L}}} + C \right]$$

$$= e^{-\frac{R}{L}t} \left[\frac{E}{R} e^{\frac{R}{L}t} + C \right]$$

$$\Rightarrow I(t) = \frac{E}{R} + C e^{-R/L t}$$

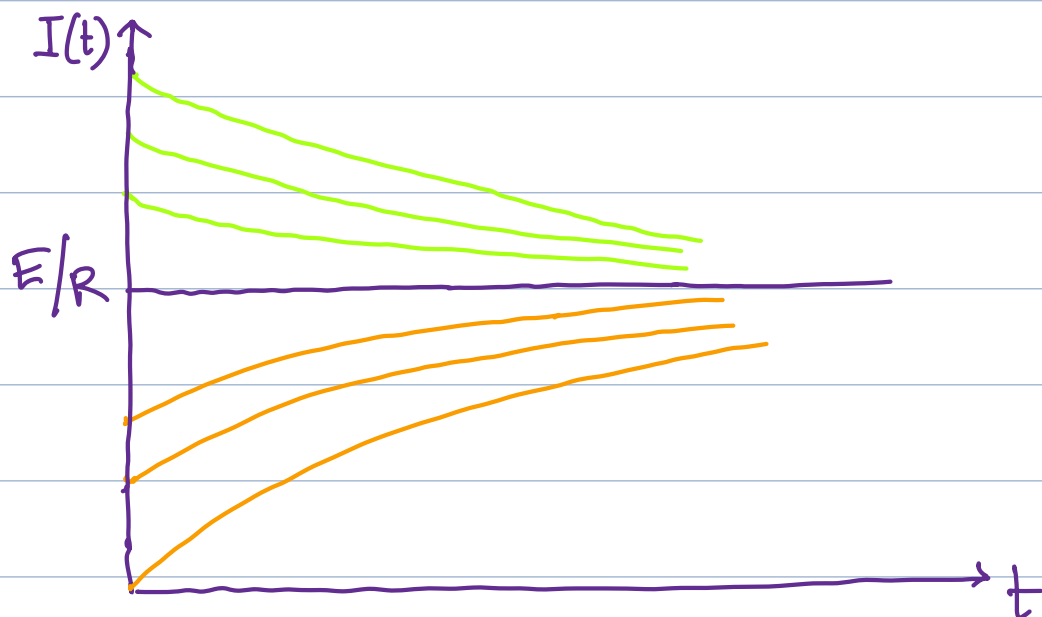
$$I(0) = 0 \Rightarrow 0 = \frac{E}{R} + C \Rightarrow C = -\frac{E}{R}$$

$$\text{So, } I(t) = \frac{E}{R} - \frac{E}{R} e^{-R/L t}$$

Given: $R = 11 \Omega$, $L = 0.1 \text{ H}$, $E = 48 \text{ V}$

$$\text{So, } I(t) = \frac{48}{11} (1 - e^{-110t})$$

$$\text{As } t \rightarrow \infty, I(t) \rightarrow \frac{48}{11} = \frac{E}{R}$$



For different initial condition, we will get different particular solution.